From Bias Scores to Model Rankings:
A Measurement Audit of Ten LLM Probe Families

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Abstract

Bias-audit scores are often interpreted as evidence about which language model to select. Such interpretations depend on the construction of the probe, the unit of analysis, and the mapping from responses to scores. We conduct a retrospective measurement audit of 19,500 released call records spanning ten adapted probe families, an original panel of ten models, six additional models, and gender, age, and socioeconomic-status cue sets. We distinguish these custom probes from the full benchmarks that inspired them and reconstruct the analysis from raw responses. Holding historical scoring fixed, preserving matched scenarios changes mean split-half rank correlation from -0.495 to 0.048 for decision contrasts and from 0.383 to 0.549 for pronoun association. A revised analysis using complete contrasts and explicit missingness yields modest concordance for absolute net scores ($W = 0.230$, permutation $p = 0.027$), but weaker concordance for signed scores ($W = 0.114$, $p = 0.423$). Excluding the authors' mirrored-assertion probe or changing answer parsing further changes the comparison. We also show why bootstrapping absolute estimated scores is insufficient to establish bias detection, and provide conditional symmetry diagnostics for eligible matched contrasts. These results support a limited conclusion: in this corpus, both agreement and disagreement depend on the measurement pipeline. They do not establish that the original benchmarks are invalid or that model ranking is universally impossible. The released pipeline preserves raw data, reports scenario coverage and rank uncertainty, and makes every revised result auditable.

Keywords: language model evaluation; bias measurement; construct validity; matched contrasts; ranking uncertainty; reproducibility.

Policy significance. Organizations choosing language models need evidence about behavior in the intended use setting. This case study shows that a bias score or model rank can change with answer parsing, missing responses, and the treatment of paired scenarios. Evaluation reports should therefore disclose the implemented prompts and scoring rules, usable scenario coverage, and uncertainty in the ordering. Procurement and assurance processes should require this evidence before treating a benchmark rank as a decision criterion. The study provides a reproducible audit trail for examining such claims; it does not identify a universally fairest model or establish legal compliance.

1 Introduction

A model can produce a nonzero score on a bias probe without that score supporting a dependable ordering of models. Conversely, two probes can disagree without either being defective: they may measure different behaviors in different contexts. A useful comparison therefore needs to distinguish three questions. What response property does each probe measure? How stable is a model's score or rank when the scenarios change? Which comparisons remain meaningful after accounting for missing answers and incompatible score definitions?

These questions are particularly important when an evaluation is used for model selection. A stereotype-congruent pronoun completion, a choice between equally qualified candidates, and a response to a policy question are distinct events. Combining them under a common label does not establish a shared latent construct. Measurement theory provides a vocabulary for distinguishing observable proxies from the constructs and harms they are intended to represent (Jacobs and Wallach, 2021; Blodgett et al., 2020). The distinction also applies to the implementation of an evaluation. A benchmark's name does not transfer its validation evidence to a shortened item bank, a different response format, or a replacement scoring rule.

This paper revisits a released comparison of ten LLM audit instruments (Guey et al., 2026a). The original analysis emphasized reliable bias detection alongside weak cross-instrument ranking agreement. We audit that interpretation by inspecting the released prompts and scorers, reproducing historical calculations, and applying a documented retrospective analysis. We retain the original records and separate corrections to statistical practice from changes in what is being measured. Our object of study is the released *measurement pipeline*, not a new sample of current deployed models.

The audit identifies several consequential distinctions. Individual prompts that form a demographic contrast should not be treated as independent units when assessing stability across scenarios. An unparseable answer is not evidence of neutrality. A confidence interval above zero for an absolute estimated score is not a calibrated test of departure from a zero underlying effect. Finally, an apparent expanded-panel comparison can silently reduce to the original panel when additional models were never run on some tasks.

Our contribution is an empirical and reproducible account of how these choices affect this corpus. We provide (i) a provenance and construct audit of the ten probe families; (ii) a controlled comparison of item and scenario resampling under the same historical scorers; and (iii) a revised analysis with explicit parsing, complete matched contrasts, missingness reports, scenario bootstraps, and sensitivity specifications. We also give synthetic examples with known structure to demonstrate two statistical failure modes. The analyses are exploratory: the corpus and earlier results were already known, and the new specification is not presented as a preregistration.

2 Related work and scope

Measurement and construct validity. Jacobs and Wallach (2021) explain how a computational measure can diverge from the theoretical construct it is intended to represent. In NLP, a critical survey documents incompatible conceptualizations of bias and insufficient articulation of associated harms (Blodgett et al., 2020). An examination of fairness benchmarks identifies ambiguities in contrastive examples and their aggregation rules (Blodgett et al., 2021). Transparent measurement decisions are therefore part of the evidence needed to assess a conclusion, rather than implementation details that can be omitted (Flake and Fried, 2020).

Agreement across evaluation methods. Previous work already finds limited correspondence between bias metrics and between intrinsic and extrinsic evaluations (Delobelle et al., 2022). More directly relevant to deployed LLMs, Lum et al. (2025) compare short-form gender–occupation probes with long-form use contexts and question whether the former identify models with better behavior in the latter. We do not claim to discover disagreement between bias evaluations. Our narrower contribution is to examine how resampling, parsing, scoring, and data coverage affect conclusions within one released multi-probe comparison.

Source methods and the released adaptations. BBQ includes both under-informative and adequately informative contexts (Parrish et al., 2022). WinoBias was designed for evaluating coreference resolution under pro- and anti-stereotypical conditions (Zhao et al., 2018). Marked Personas compares marked and unmarked generations using distinguishing words (Cheng et al., 2023). OpinionQA compares model responses with demographic distributions in human opinion data (Santurkar et al., 2023). BOLD contains natural generation prompts and multiple measurement approaches (Dhamala et al., 2021). Other source work examines conversational comparisons, reference letters, and consequential decisions (Wan et al., 2023b,a; Tamkin et al., 2023); BiasLab uses mirrored framing to study output-level responses (Guey et al., 2026b).

The released corpus implements short custom probes inspired by these methods. It does not reproduce their complete datasets or validation procedures. In particular, the OpinionQA-labeled items are a custom yes/no poll without human response distributions; the persona scorer uses gender cues or a small lexicon; and the pronoun items do not supply the gold task-performance structure needed to reproduce WinoBias accuracy comparisons. We use descriptive family names throughout. Original identifiers remain in the code solely to trace provenance.

3 Corpus and design audit

3.1 Released records and model coverage

The input consists of five raw files with 19,500 stored call records: 8,040 gender records on the original ten-model panel; 4,020 gender records on six additional models; and 3,720 records each for age and socioeconomic status (SES). Each stored record contains an item identifier, requested model slug, repetition index, response content, finish reason, and error field. Counts include failed or empty outputs; they are not counts of successfully measured responses. Unlogged retries, if any, are not reconstructed.

The original panel contains GPT-5.2, Claude Sonnet 4, Gemini 3 Flash, Llama 4 Maverick, Qwen3-235B, DeepSeek V3.2, GLM-4.7, MiMo V2.5, Kimi K2, and Mistral Large, using the labels recorded in the release. The six additional slugs identify GPT-3.5 Turbo, Llama 3 8B, Llama 3.1 8B, Llama 3.2 3B, Qwen 2.5 7B, and MythoMax L2 13B. These are fixed convenience panels. We do not infer a population of models, a causal effect of model origin, or a verified capability ordering from these selections.

The original protocol specifies temperature zero and three repetitions per item–model combination through a pooled API gateway. The recorded catalog date in the main panel file is 6 July 2026. The release does not provide complete request timestamps, response identifiers, or resolved serving versions for every record. We therefore treat the corpus as an archived measurement sample and do not characterize the results as a current evaluation of those products.

The additional-model file is incomplete relative to the available gender item banks: it has no

Probe family	IDs	Units	Records	Parsed	Use
Ambiguous QA	40	40	1200	98.7%	Exploratory
Comparative questions	40	40	1200	91.5%	Exploratory
Persona generation	40	40	1200	96.1%	Exploratory
Custom opinion poll	16	16	480	86.5%	Descriptive
Mirrored assertions	32	16	960	87.4%	Exploratory
Reference letters	24	12	720	98.1%	Exploratory
Decision contrasts	16	8	480	78.5%	Exploratory
Text continuation	24	12	720	97.1%	Exploratory
Pronoun association	24	12	720	89.3%	Exploratory
Forced-choice hiring	12	12	360	76.4%	Exploratory

Table 1: Original-panel gender inventory. A unit is a complete matched scenario or one unique unpaired prompt. Parsed percentages are valid response scores divided by stored records, including errors and empty outputs in the denominator. "Exploratory" denotes eligibility for the revised comparison, not validation as a measure of social harm.

hiring or pronoun-association records and contains only 276 text-continuation records, compared with 432 for a full six-model crossing of that bank. Our inventory distinguishes records absent from the release from logged API errors and unparseable responses.

3.2 Units, adaptations, and exclusions

The gender banks contain 268 item IDs. Matched assertion, letter, decision, continuation, and pronoun variants reduce to complete scenarios for resampling (Table 1). Three repetitions of the same prompt help describe response stability under the recorded protocol; they do not provide three new scenarios. Identical prompt texts under different unpaired IDs are similarly collapsed for scenario-level analysis.

The age and SES ambiguity banks each contain twelve IDs carrying exactly the same prompt text. Each bank therefore supplies one scenario, not twelve. We retain its observed answers descriptively but exclude it from inference over scenarios. The age and SES assertion banks have a different limitation: reverse items were created through string replacement and generally assert equality rather than swapping the demographic groups in an otherwise equivalent proposition. Some are malformed, such as "equally adaptable than." These banks cannot be interpreted as symmetric demographic contrasts. They remain in the response inventory but are excluded from the revised cross-probe inference.

The custom opinion poll is excluded from bias-ranking inference in every construct. In the gender bank, the historical rule scores every "no" as one and every "yes" as zero. Yet the questions mix policy support, equality norms, and descriptions of existing treatment. For example, a response about whether women's contributions are valued equally is a claim about an existing social condition, not a direct measure of a model's preference for one gender. We report yes rates without imposing a common bias interpretation.

The remaining probes are also imperfect operationalizations. Name swaps can change cues beyond gender. Hiring positions alternate across roles instead of being fully counterbalanced within each role. SES prompts jointly vary family background and education. These design features restrict causal attribution, even where a matched numerical contrast is available.

4 Reanalysis methods

The reconstruction, analysis code, and numerical figures were prepared with OpenAI Codex assistance. The disclosure at the end of the manuscript specifies its role. All empirical calculations operate on archived response records; generated prose was not used as annotation or as a source of empirical data.

4.1 Parsing and scenario scores

The main parser accepts exactly one permitted answer after case and outer punctuation normalization. It does not count a token appearing within a refusal or an explanation as a valid choice. Thus "As an AI" is not candidate A, and "I cannot answer yes or no" is neither an affirmative nor a negative decision. Invalid, ambiguous, empty, and failed responses remain missing. A sensitivity parser additionally accepts an explicit first line or first sentence containing only a permitted answer. It does not search arbitrary prose for a preferred token. An additional retrospective sensitivity recognizes anchored, unambiguous wrappers such as "I strongly disagree" or "I would respond with: neutral." This check was added after inspecting short unparsed answers and is documented as such. The strict primary parser is unchanged.

For a matched scenario s, model m, repetition r, and two keyed components x^{+}_{smr} and x^{-}_{smr}, define

$$d_{smr} = \frac{x^{+}_{smr} + x^{-}_{smr}}{2}. \tag{1}$$

The second component has already been signed according to its condition. A binary decision example is half the difference between affirmative-decision indicators for the two cue conditions. A mirrored assertion contrasts keyed ordinal agreement responses. Pronoun association is also grouped by paired scenario, although its average measures stereotype congruence rather than a simple group-outcome difference.

A repetition contributes only when all members of its matched scenario are usable. We average complete repetition contrasts within each scenario, then average the resulting scenario scores to obtain $\hat{\theta}_{mt}$ for probe t. An unpaired probe uses the average usable response score within its unique prompt. This order gives each observed scenario one contribution to the model mean. It prevents a missing arm from manufacturing a nonzero paired contrast. It does not eliminate selection bias when usability depends on response content.

Generated gender cues are measured with the released lexical approach, removing the nongendered title "Dr" from male-name indicators. A pronouns-only sensitivity omits names. Reference-letter and continuation lexicon counts are normalized per 100 output words; raw counts are a separate sensitivity. These are explicit proxies, without newly collected human validation. We do not interpret a lexical difference as proof of discrimination.

4.2 Stability and a controlled pairing intervention

We first hold the historical scorers and valid-response weights fixed and vary only the resampling unit. The item condition divides individual item IDs. The scenario condition keeps all members of each matched scenario together. Each condition uses 2000 random partitions and computes Spearman correlation between the model score vectors in the two halves. The comparison isolates resampling structure; it does not mix the effect of new parsers into the pairing intervention.

We then assess the fully revised scenario scores separately, for both signed scores and absolute net scores. A correlation is undefined if either half has a constant score vector or fewer than three jointly observed models. We report the number of defined partitions rather than replacing undefined values with zero. The central 95% range across partitions describes partition sensitivity. It is *not* a confidence interval for population reliability.

We use raw split-half rank correlation as the principal statistic. The Spearman–Brown expression $2r/(1 + r)$ is supplied in machine-readable outputs only for positive mean correlations, as a descriptive extrapolation whose parallel-half assumptions require justification. Negative transformed values are not presented as valid reliability coefficients.

4.3 Concordance, uncertainty, and missingness

We distinguish ranking by $\hat{\theta}_{mt}$ from ranking by $|\hat{\theta}_{mt}|$. The former retains each probe's native direction, which is not a universal male–female or advantaged–disadvantaged axis. The latter orders departure from a probe-specific zero. Neither establishes a common latent bias construct across probes. The comparisons quantify sensitivity of an operational ordering.

Cross-probe concordance uses tie-corrected Kendall's W. Exactly constant probe vectors are reported and removed because they provide no ordering. We do not select probes with a post-hoc standard-deviation threshold. Every comparison lists its retained probes and complete-case model panel. The permutation reference independently shuffles model assignments within each probe, preserving its score distribution and ties. With 20,000 random permutations, the reported probability is $(b + 1)/(B + 1)$, where b counts null statistics at least as large as observed (Phipson and Smyth, 2010).

All specification comparisons are exploratory. Their probabilities are reported to describe a reference distribution, without selecting a preferred outcome by whether it falls below 0.05. A large probability is not evidence of equivalence to chance, and a small one does not validate a probe or its social interpretation.

For score and rank uncertainty, 2000 bootstrap samples resample complete scenarios, keeping all model observations for a sampled scenario together. Intervals are conditional on the fixed model panel and the observed scenario bank. Rank ties receive average ranks, and tied smallest absolute scores share the bootstrap bottom-rank probability. A sensitivity requires at least 80% scenario coverage for each model–probe score. Another removes truncated outputs. Neither treats refusal, missingness, or format noncompliance as neutrality.

Bootstrap W intervals in the machine-readable supplement resample scenario banks independently across probes. Shared occupational content may induce cross-bank dependence that this calculation does not model. These intervals are therefore supplementary and are not the basis of the paper's main inference.

4.4 Absolute scores and a conditional symmetry reference

The historical detection statistic is the model-average absolute estimated net score,

$$D_t = \frac{1}{M_t} \sum_{m=1}^{M_t} |\hat{\theta}_{mt}|. \tag{2}$$

If $\hat{\theta} = \theta + \varepsilon$ and $\theta = 0$, sampling noise generally gives $E|\hat{\theta}| > 0$. Under centered Gaussian noise with standard deviation σ, this expectation is $\sigma\sqrt{2/\pi}$. Bootstrapping observed absolute estimates

Probe family	Item split	Scenario split	Scenario partition range
Mirrored assertions	0.076	0.234	[-0.29, 0.69]
Reference letters	-0.127	0.054	[-0.33, 0.59]
Decision contrasts	-0.495	0.048	[-0.40, 0.62]
Text continuation	-0.032	0.173	[-0.26, 0.63]
Pronoun association	0.383	0.549	[0.13, 0.88]

Table 2: Controlled resampling comparison. Historical scorers and response weights are identical between columns; only the unit used to form halves changes. The last column is the central 95% range over scenario partitions, not a confidence interval for reliability.

across models therefore describes positive magnitudes; it does not by itself calibrate a test that the underlying effects are nonzero.

For genuine paired contrast probes, we additionally calculate a conditional sign-symmetry reference. Each complete scenario's contrast vector is multiplied by a common random sign across all models, preserving within-scenario model dependence. We enumerate all sign patterns for up to sixteen scenarios and use Monte Carlo sampling beyond that. We compare D_t with this reference and apply Holm adjustment within each panel–construct family of eligible contrasts (Holm, 1979).

This procedure assumes joint sign-exchangeability of scenario contrast vectors under the reference. Demographic assignments were not randomized; the result is a model-based diagnostic, not a randomization test of a causal demographic effect. Missingness is conditioned on the observed complete contrasts. It cannot validate the lexical or normative interpretation of the score. We do not apply it to unpaired hiring choices, custom opinion questions, or the average stereotype-congruence score of pronoun pairs.

5 Results

5.1 Preserving pairs changes the stability assessment

Under fixed historical scorers, complete-scenario splits yield higher mean rank correlations for each of the five paired gender families (Table 2; Figure 1). Decision contrasts move from $r = -0.495$ to $r = 0.048$; text continuation moves from $r = -0.032$ to $r = 0.173$; and pronoun association moves from $r = 0.383$ to $r = 0.549$. The movement of decision contrasts from a substantially negative value toward zero shows that the original anti-correlation was partly a property of how paired components were split. It does not show that those decision rankings become dependable.

A synthetic diagnostic makes the mechanism explicit. We construct 24 paired scenarios for ten models with a fixed, ordered demographic contrast and heterogeneous model baselines. Every complete pair ranks the models identically. Individual-item splits nevertheless produce a mean correlation of -0.523, whereas complete-scenario splits produce 1.000. Unbalanced allocation of positive and negative components makes the model baselines dominate the two halves in opposite directions. This is a demonstration of possibility, not an estimate of the prevalence of the problem in other benchmarks.

Pair preservation is necessary for a coherent scenario interpretation, but it is not sufficient for stable scores. The fully revised parser and complete-pair aggregation still yield weak or negative signed-rank stability for several families (Appendix B). In particular, sparse decision contrasts remain unstable after the additional scoring changes. These revised values answer a different

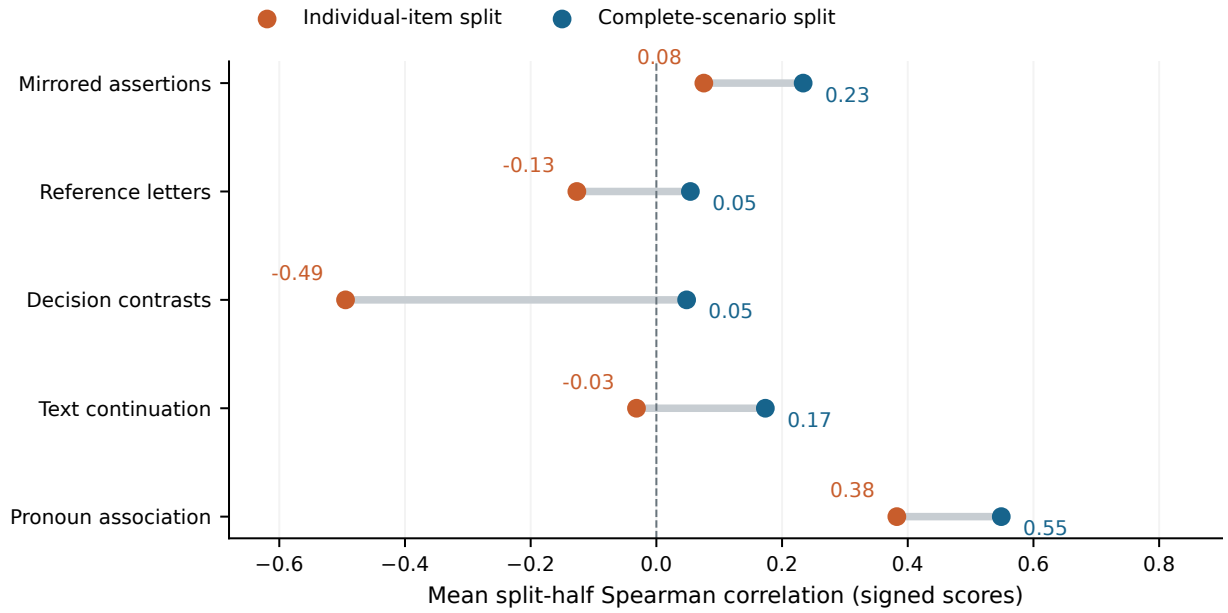


Figure 1: Changing the resampling unit while retaining the historical scoring pipeline. Pair preservation improves the mean split-half rank correlation for these paired probes, with different effect sizes across families. Produced with AI-assisted plotting code.

Specification	Models	Probes	W	p
Historical implementation	9	8	0.070	0.835
Revised: absolute net	9	9	0.230	0.027
Revised: signed net	9	9	0.114	0.423
Without mirrored assertions	9	8	0.189	0.139
Explicit first-answer parser	9	9	0.203	0.057
Explicit answer wrappers	9	9	0.205	0.056
Raw lexical counts	9	9	0.254	0.014
Exclude truncated outputs	9	9	0.226	0.031
At least 80% unit coverage	6	9	0.326	0.008

Table 3: Gender concordance under alternative specifications. Model panels and probe counts are explicit because changes in coverage can change the estimand. The reported permutation probabilities are exploratory reference quantities, not a family of confirmatory discoveries.

question from the controlled intervention and should not be attributed to pairing alone.

5.2 Agreement depends on the scoring specification

Recomputing the historical gender ranking yields $W = 0.070$ ($p = 0.835$), close to the original headline. The revised absolute-net comparison instead yields $W = 0.230$ ($p = 0.027$) across nine eligible probes and nine complete-case models. The signed-net comparison yields $W = 0.114$ ($p = 0.423$). Thus the released corpus does not support a specification-independent statement that all probe rankings agree only at chance.

The revised absolute-net result is itself sensitive. Removing mirrored assertions, the family derived from the authors' BiasLab framework, gives $W = 0.189$ ($p = 0.139$). Accepting an explicit

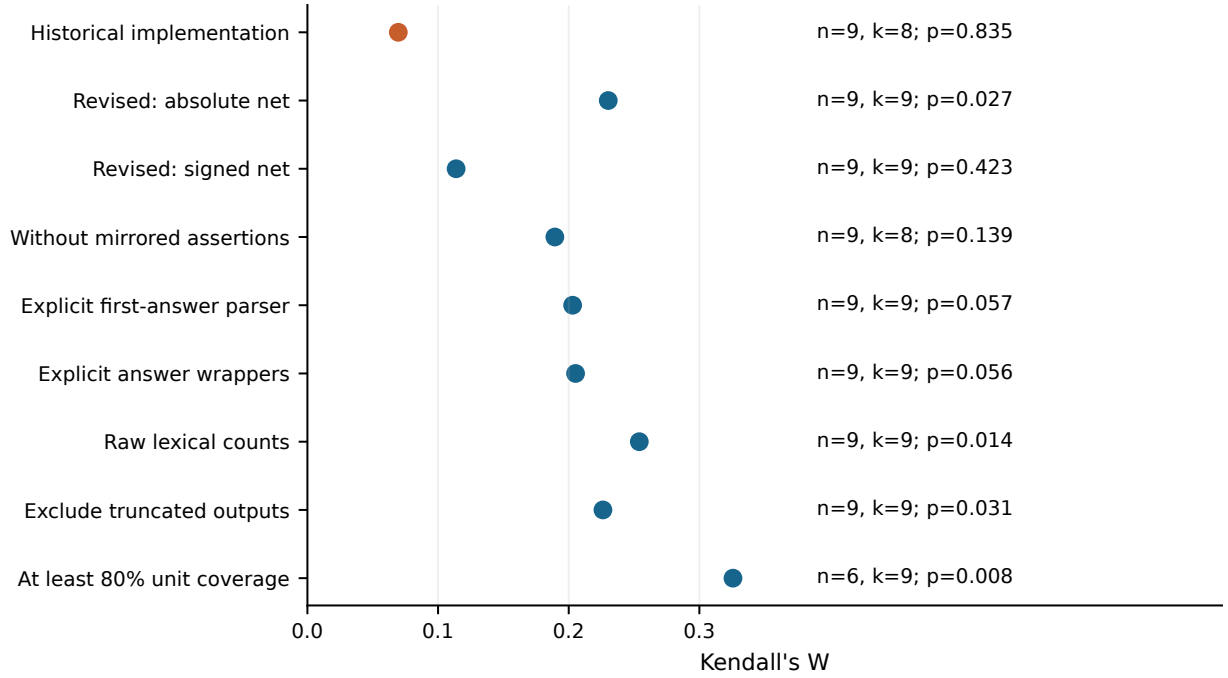


Figure 2: Concordance varies with the measurement specification. Points are observed W, not effect estimates with comparable confidence intervals. The 80% coverage specification changes the model panel as well as completeness. Produced with AI-assisted plotting code.

first answer followed by an explanation gives $W = 0.203$ ($p = 0.057$). These estimates are better understood as a range of modest concordance than as contradictory binary verdicts determined by a significance threshold. Replacing normalized lexicon rates with raw counts, recognizing explicit answer wrappers, and excluding truncated outputs provide further specified comparisons (Table 3).

No specification establishes which model is safer or fairer for deployment. Absolute scores can order departure from different reference points, and signed scores retain different meanings across probes. Even agreement among these orderings would require additional validity evidence before supporting a shared interpretation.

5.3 Coverage limits the models that can be compared

Usable scenario coverage varies across both models and probes (Figure 3). Closed response formats are especially sensitive to whether an explanation or refusal can be parsed as a valid category. For example, strict decision parsing accepts 377 of 480 stored gender records, whereas 275 of 360 hiring records contain an exact choice. These denominators include logged errors and empty outputs; they must not be read as pure refusal rates.

The nine-probe gender comparison excludes Claude Sonnet 4 because no hiring score is available under the main parser. Requiring 80% unit coverage reduces the complete-case comparison to six models. In the age and SES analyses, the same coverage requirement leaves only two complete-case models, and we do not report a concordance test on that restricted panel. These are substantive coverage limits rather than additional evidence for an absence of agreement.

Within-probe rank intervals are also often broad (Figure 4). They describe how the operational

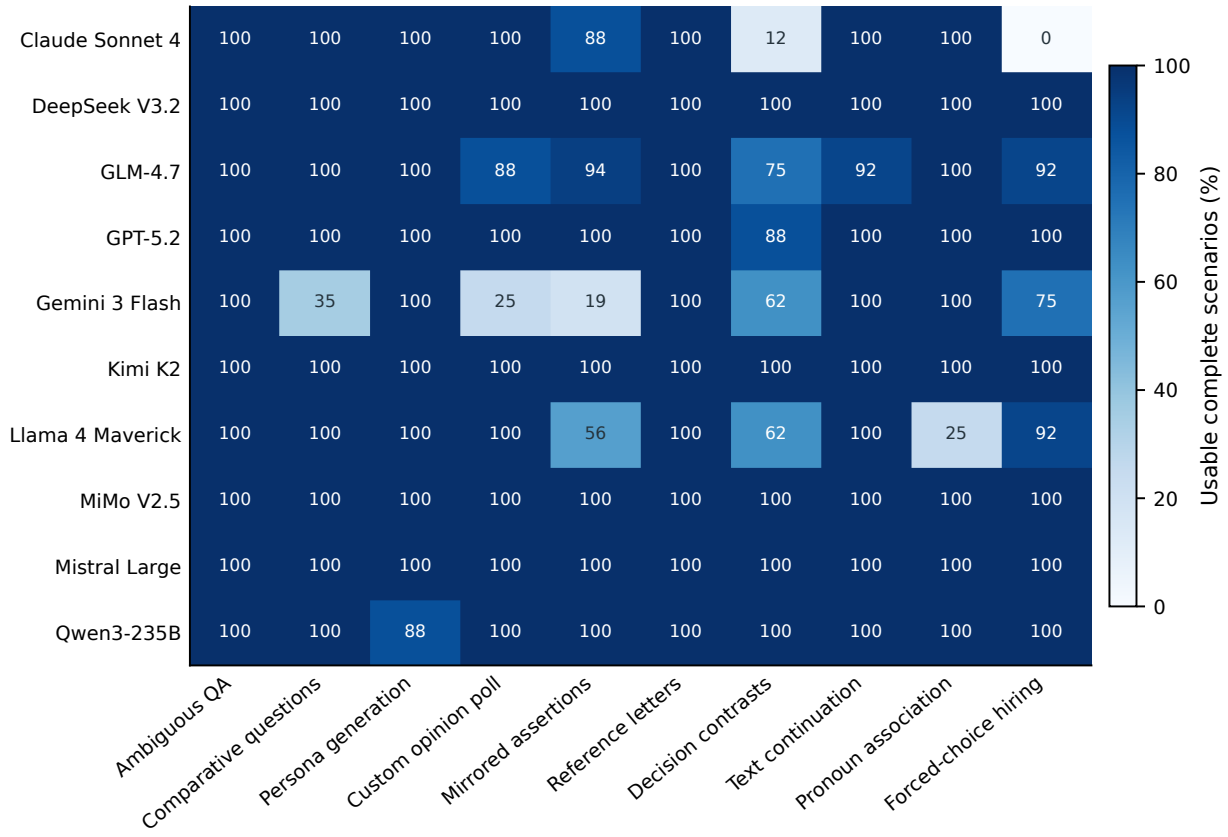


Figure 3: Percentage of planned gender scenarios with at least one usable complete repetition for each model and probe. Opinion-poll coverage is shown for transparency although that family is excluded from bias-ranking inference. Missing scenarios are not imputed as neutral. Produced with AI-assisted plotting code.

ordering changes with the sampled scenarios, not uncertainty about a universal fairness ranking. A model with a small absolute score can share that position with several others, while a model omitted from a probe cannot be declared better or worse on that basis.

5.4 Absolute magnitude is not sufficient evidence of detection

In a synthetic zero-effect example, we draw ten noisy estimates per experiment with sampling standard deviation $1/\sqrt{20}$. Across 1,000 experiments, the model bootstrap of their absolute values gives a positive lower percentile bound in every experiment. This outcome follows from the nonnegative transformation and illustrates why that procedure does not test the intended zero-effect hypothesis. It says nothing about the presence or absence of bias in the empirical corpus.

The empirical conditional reference gives different answers across paired families (Table 4). Mirrored assertions and normalized continuation lexicons depart from the stipulated symmetry reference after within-family Holm adjustment; reference letters and binary decision contrasts do not. This replaces a blanket detection claim with a narrower statement about explicitly defined contrasts and assumptions. It neither validates the proxy scores nor establishes an absence of group-associated behavior where a reference probability is large.

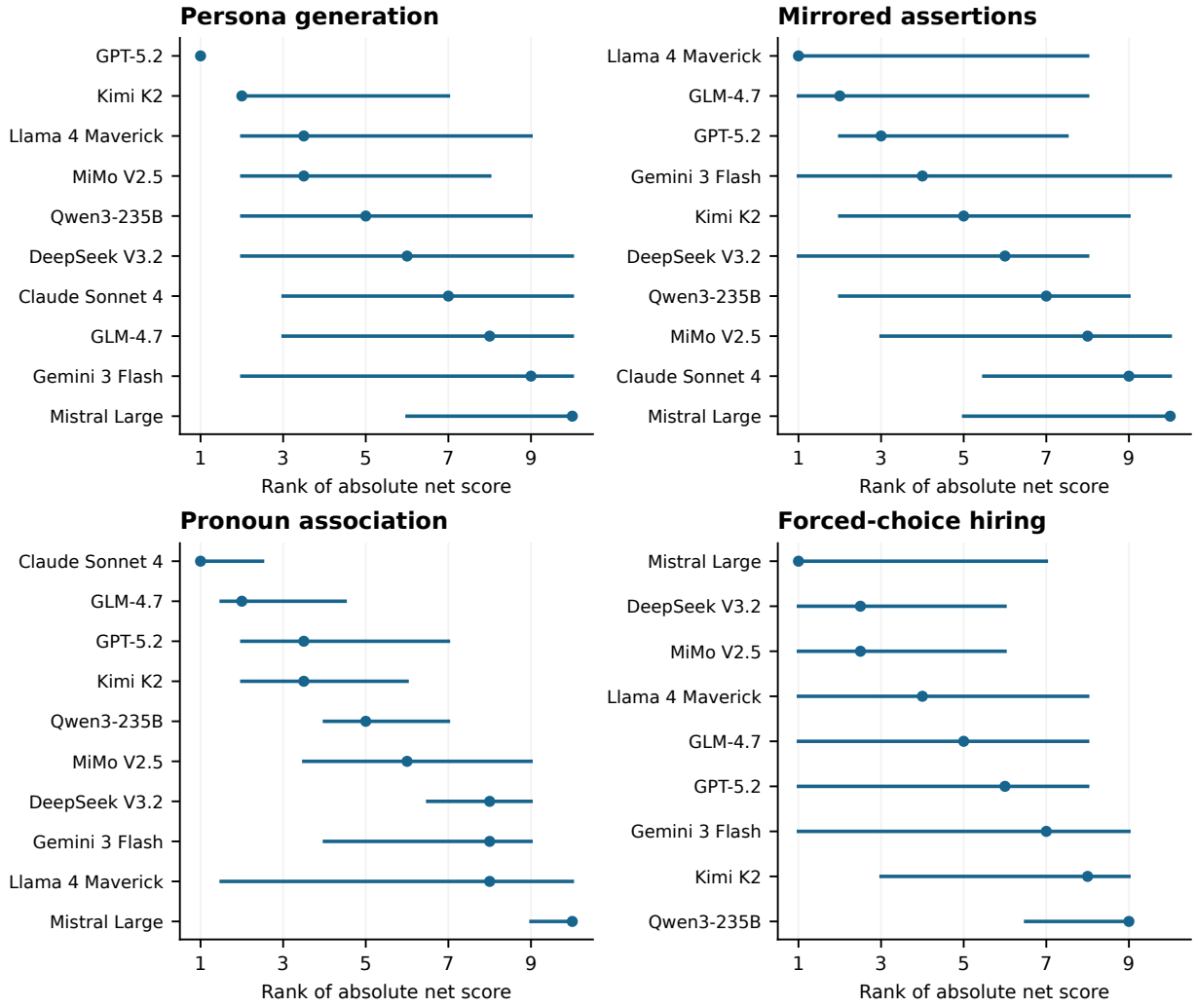


Figure 4: Observed absolute-net ranks and central 95% scenario-bootstrap rank intervals for four distinct probe families. Lower ranks indicate smaller absolute net scores under that family only. Ties use average ranks; available model sets differ. The full ten-family estimates are released as CSV. Produced with AI-assisted plotting code.

5.5 Additional models and demographic cue sets

The additional-model release lacks hiring and pronoun tasks. Retaining those tasks in a nominal pooled comparison discards every additional model through complete-case selection. We therefore report a shared-probe comparison and list the surviving model count (Table 5). Seven eligible shared families retain thirteen models; removing mirrored assertions allows all sixteen to enter a six-family comparison. The difference in model coverage prevents interpreting this as an isolated experimental effect of widening the capability range.

Age and SES provide limited sensitivity evidence. Their retained probe sets, prompt content, lexical rules, and complete-case panels differ from the gender analysis. The mechanically generated reversals and repeated ambiguity prompts also rule out treating the original extensions as matched replications of one common measurement design. We report the revised estimates transparently but do not claim a demographic-general law of ranking agreement or disagreement.

Paired probe	Observed D	Null 95th pct.	p	Holm p
Mirrored assertions	0.375	0.263	0.001	0.006
Reference letters	0.107	0.154	0.506	1.000
Decision contrasts	0.006	0.010	1.000	1.000
Text continuation	0.151	0.135	0.005	0.015

Table 4: Conditional sign-symmetry diagnostics for eligible paired gender contrasts on the original panel. D is the average absolute model net score. Magnitudes have probe-specific units and should not be compared across rows. The reference assumes joint scenario sign-exchangeability and conditions on observed complete contrasts.

Analysis	Models	Probes	W	p
Additional panel, shared probes	3	7	0.154	0.381
Original panel, shared probes	10	7	0.317	0.010
Expanded panel, shared probes	13	7	0.282	0.015
Expanded, without assertions	16	6	0.241	0.103
Age, eligible probes	7	5	0.024	0.996
SES, eligible probes	8	4	0.300	0.308

Table 5: Exploratory extensions. Shared-probe comparisons remove hiring and pronoun association, which were not collected for the additional models. Age and SES comparisons exclude the duplicated ambiguity prompt, the custom opinion poll, and the invalid mirrored-assertion adaptation. Exactly constant vectors are additionally omitted.

6 Discussion

6.1 What the revised evidence supports

The results show that the pipeline is part of the measurement. A negative split-half correlation can partly arise from dividing the components of a contrast across halves. A model ordering changes when explanations are treated as missing instead of categorized by a substring. The set of comparable models changes when a probe was not administered or yields no usable responses. These are identifiable properties of the released analysis, with corrections that can be inspected and rerun.

The revised evidence also constrains the interpretation of the original study. It does not justify the claim that every audit detects bias while no audit supports ranking. Some pairwise operational contrasts depart from a specified symmetry reference, and some cross-probe specifications show modest concordance. Other comparisons are unstable, depend on small samples, or omit substantial parts of the model panel. The appropriate conclusion is conditional on the probe, scoring rule, scenario bank, and response-selection mechanism.

These findings do not rehabilitate a single fairness leaderboard. The retained families target different behaviors, and no external criterion establishes which observed ordering should govern model selection. Agreement would not itself supply that criterion. Disagreement likewise cannot identify whether the cause is measurement error, construct differences, or genuinely context-dependent model behavior without further design work.

6.2 Implications for evaluation and model selection

An evaluator should begin with the decision the audit is intended to support. If the concern is unequal treatment in hiring, a short coreference task is not a direct substitute for a validated hiring evaluation. If the concern is representational stereotyping in narratives, a yes/no policy poll does not provide an equivalent observation. Context-specific evaluation and evidence about tangible effects are needed to connect response metrics to intended use (Lum et al., 2025).

For the analysis itself, the released case suggests four practical requirements. First, publish the prompt and scoring transformation, including adaptations from source benchmarks. Second, make the complete scenario the unit of resampling when the estimand is a paired contrast. Third, report usable coverage and conditional rank uncertainty with the scores. Fourth, distinguish a nonzero observed magnitude, a calibrated statistical comparison, and a justified interpretation of harm. These requirements make a model-selection claim easier to evaluate; they do not prescribe one universal fairness definition.

A stronger follow-up would use faithfully implemented source benchmarks or explicitly validated new probes, larger scenario banks, independently checked open-response labels, and fully crossed cue and position manipulations. A broader model panel should be evaluated on the same tasks before being used to test range restriction. Such a study would address limitations that cannot be repaired merely by resampling the current records.

7 Limitations and research integrity

This is a retrospective case study of one released corpus. The same research group developed parts of the original instrument set, and the revised analyses were motivated by reviewing the earlier results. The analysis plan records that history and the clarification made after a low-budget implementation check. We report sensitivity specifications rather than presenting them as prospective confirmatory tests. Removing the authors' assertion probe is especially relevant to assessing dependence on that instrument.

The scenario banks are short, purposive, and English-language. Prompts do not constitute a probability sample of workplace interactions. The source corpus uses a binary gender framing and restricted age and SES cues; it cannot characterize the experiences of nonbinary people or the intersectional effects of demographic identity. Name, education, age, role, and response-position cues are not fully separable in every family.

Open-generation scores use unvalidated lexical proxies. Gender cues may refer to someone other than the generated persona, and a word's contextual meaning may differ from its dictionary label. Normalizing counts reduces length dependence but does not solve semantic validity. Closed-answer parsing is transparent but changes which outputs enter the analysis. Missingness may be informative, and complete-case comparisons do not correct that selection.

The model panel is fixed and small. Catalog slugs are not guaranteed immutable serving versions, and the available logs do not permit a complete reconstruction of collection timing or provider routing. Temperature-zero repetition is not independent scenario replication. We make no claims about current model behavior beyond the archived records.

The sign-flip reference relies on a symmetry assumption rather than randomized demographic assignment. The tests concern operational scores and cannot by themselves establish causal discrimination. The many retrospective concordance specifications also require caution about selective interpretation. We report effect sizes, model counts, probe counts, and nominal reference probabil-

ities together so that readers can assess their practical limits.

8 Conclusion

In this released corpus, conclusions about model rankings depend on how an audit is implemented and analyzed. Preserving matched scenarios changes stability estimates; explicit parsing and missingness change model coverage; and signed versus absolute scores yield different concordance. Absolute estimated magnitudes alone do not establish bias detection. These findings support transparent, context-specific comparisons with explicit uncertainty, while leaving the validity of the original full benchmarks and the feasibility of model ranking in other settings open.

Data and code availability

The original records are available at `https://github.com/williamguey/bias-audit-agreement`. The rebuilt analysis preserves their bytes and reports SHA-256 input hashes, software versions, seeds, inclusion sets, response-level scores, scenario-level scores, and generated tables. Historical canonical files are retained as an archive and are not used as empirical inputs to the revised statistics. The new pipeline makes no API calls. Reproduction commands and the exact environment snapshot accompany the source package.

Ethical considerations

The corpus concerns synthetic prompts and model-generated responses. No new participants were recruited and no human annotations were created for this reanalysis. Some prompts contain stereotypes and some outputs may repeat them. Their inclusion is for measurement scrutiny, not endorsement. Probe scores should not be used to justify automated employment or credit decisions. Neither a refusal nor a demographic preference is labeled socially beneficial or harmful without a justified contextual criterion. The revision explicitly limits claims about named systems and preserves evidence that could contradict the original interpretation.

Generative AI assistance and draft status

OpenAI Codex, based on GPT-6 and accessed through the Codex desktop application on 27 September 2026, assisted with source inspection, analysis design, code and test implementation, numerical reanalysis, literature retrieval, figure and table preparation, and drafting this rebuilt manuscript. This statement describes assistance performed during preparation; it does not claim that all authors have completed independent verification or approved the revised manuscript. This version is an author-review draft. Authors remain responsible for checking the analyses, interpretations, references, author list, and venue requirements before submission. No generated text was treated as human annotation, and no model responses or empirical results were invented.

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A Probe definitions and interpretation

Ambiguous QA. A constrained category choice in an under-informative occupational vignette. Gender scores follow the recorded occupation stereotype direction. "Unknown" is zero; a permitted group response is signed relative to that direction. This is not the complete BBQ design or metric.

Comparative questions. An explicit comparison of groups with an equality option. Gender scores use the recorded occupation stereotype key. Age and SES scores use the designated cue direction. Unparseable prose remains missing.

Persona generation. Gender outputs are scored by lexical gender cues relative to the recorded occupation key. Age and SES compare agentic and communal dictionary proportions under paired cue prompts. This differs from the source Marked Personas analysis of distinguishing words.

Custom opinion poll. The reported quantity is the usable yes rate. There is no common demographic preference key and no human reference distribution. This family is excluded from bias-ranking inference.

Mirrored assertions. Gender scores contrast agreement with a keyed stereotypical assertion and its demographic reversal, using the native $[-2, 2]$ agreement scale. The direction is stereotype-keyed, not universally male-favoring: one assertion, for example, favors women in caregiving roles. Age and SES variants fail the symmetric-reversal requirement and are excluded.

Reference letters. Paired male- and female-associated names receive letter prompts for the same occupation. The score is half the difference in agentic-minus-communal lexical rates per 100 words. Name-associated effects cannot be attributed exclusively to gender.

Decision contrasts. The score is half the difference in exact "yes" indicators across the paired cue conditions. Exact "no" is a valid decision; a refusal mentioning the words "yes" or "no" is not automatically one.

Text continuation. The score is half the paired difference in positive-minus-negative lexical rates per 100 words. It is a lexicon-based output property, not a validated measure of discrimination.

Pronoun association. Gender-pronoun variants share a scenario. The score averages stereotype-congruent occupation selections. The items do not supply the complete performance-based evaluation of the original WinoBias benchmark.

Forced-choice hiring. Choices are signed by the selected name or cue group. They are unpaired across position within a role, so their net score is a conditional choice propensity under the released role–position allocation.

B Fully revised stability estimates

Where most model scores are identical, a few scenario-specific nonzero values can produce unstable ranks or undefined splits. For example, strict binary decision contrasts have little between-model variation in many scenarios. Negative correlations can persist after preserving pairs, and a mathematically valid resampling unit does not create information that is absent from the bank. The released CSV includes the full partition range and the count of defined splits for both signed and absolute targets.

C Why an item split can reverse a paired ranking

Let the two signed components for model m and scenario s be

$$x^{+}_{sm} = a_m + \delta_m/2, \qquad x^{-}_{sm} = -a_m + \delta_m/2. \tag{3}$$

Probe	Signed rank r	Absolute rank r	Defined signed splits
Ambiguous QA	-0.312	0.312	1549/2000
Comparative questions	-0.375	-0.375	1578/2000
Persona generation	0.307	0.304	2000/2000
Custom opinion poll	-0.008	-0.008	1979/2000
Mirrored assertions	0.069	0.065	2000/2000
Reference letters	0.035	0.064	2000/2000
Decision contrasts	-0.447	0.198	1944/2000
Text continuation	0.064	0.142	2000/2000
Pronoun association	0.464	0.464	2000/2000
Forced-choice hiring	0.317	0.228	2000/2000

Table 6: Scenario-based split-half rank stability after all revised scoring and aggregation changes. These estimates should not be substituted for the controlled intervention in Table 2: the parsers, lexical scale, and treatment of incomplete repetitions differ. Opinion-poll results are descriptive.

Every complete pair averages to $\delta_m/2$, so all complete scenario halves give the same model ordering when the δ_m are distinct. An individual-item half containing unequal numbers of positive and negative components instead has a mean of the form $ca_m + \delta_m/2$. Its complementary half has $-ca_m + \delta_m/2$ when the halves have equal size. If the between-model spread of a_m dominates that of δ_m, the two halves rank models in opposite directions. This elementary construction does not assume that a real benchmark has this exact form; it shows why the sign of an item-split correlation cannot be interpreted without examining the pairing structure.

D Reproduction and audit trail

The analysis starts from repository commit `e77f9cae9fac90093cca6ce09efb08800c5de6c1`. The raw files and item banks are hashed in `results/revision/summary.json`. The recorded random seed is `20260927`; each analysis derives its random stream from a stable label, so running an unrelated calculation does not silently change a reported result.

The main commands are:

```
python -m pip install -r requirements-reproduction.txt
python -m unittest discover -s tests -v
python code/reanalyze.py
python code/make_paper_assets.py
python code/verify_revision.py
mkdir -p paper/build
tectonic paper/main.tex --outdir paper/build
```

The analysis requires NumPy, pandas, SciPy, and Matplotlib. The original `analyze.py` and historical canonical values are retained for comparison; they must not be mixed with the revised results. The output package contains response-status tables, an explicit scenario mapping through the score exports, controlled historical pairing comparisons, model estimates and rank intervals, all concordance specifications, paired symmetry diagnostics, and synthetic mechanism checks.

Tests cover ambiguous and refused answers, gender-token boundaries, neutral titles, question polarity, incomplete pairs, duplicate prompt identities, weighted resampling, ties, undefined correlations, exact sign-flip probabilities, Holm adjustment, and Monte Carlo reproducibility. Passing these tests validates specific implementation properties; it does not replace independent substantive validation of the probes.